MTH166

Lecture-13

Solution of Non-Homogeneous LDE with Constant Coefficients Using Operator Method-I

Topic:

Solution of Non-Homogeneous LDE with Constant coefficients Using Operator Method-I

Learning Outcomes:

Solving Non-Homogeneous LDE Using operator method when:

1. Function is of the form: $r(x) = e^{\alpha x}$

2. Function is of the form: $r(x) = \cos \alpha x$ or $r(x) = \sin \alpha x$

Solution of Non-homogeneous LDE with constant coefficients Using Operator Method:

Let us consider 2nd order Non-homogeneous LDE with constant coefficients as:

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = r(x) \qquad (1)$$

or

$$ay'' + by' + cy = r(x) \qquad (1)$$

Let $\frac{\boldsymbol{d}}{\boldsymbol{dx}} \equiv \boldsymbol{D}$ be Differential operator (An algebraic operator like $+, -, \times, \div$)

Equation (1) becomes:

$$aD^2y + bDy + cy = r(x)$$

Symbolic Form (S.F.): $(aD^2 + bD + c)y = r(x)$

$$\Rightarrow f(D)y = r(x)$$

To find Complimentary Function (C.F.):

A.E.: $f(D) = 0$

$\Rightarrow \ (aD^2 + bD + c) = 0$

$\Rightarrow D = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = m_1, m_2$ (Say) (Suppose here $m_1 \neq m_2$ are real roots)

Complementary function C.F. is given by:

$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$

To find Particular Integral (P.I.):

P.I.: $y_p = \frac{1}{f(D)} r(x)$ (There are different methods to evaluate it)

General Solution: $y =$ C.F. +P.I.

i.e. $\boldsymbol{y = y_c + y_p}$

Operator Method to find Particular Integral (P.I.):

Case 1: If $r(x) = e^{\alpha x}$, then P.I. $y_p = \frac{1}{f(D)} r(x)$

$\Rightarrow y_p = \frac{1}{f(D)} e^{\alpha x} = \frac{1}{f(\alpha)} e^{\alpha x}$, (i.e. Put $D = \alpha$), provided $f(\alpha) \neq 0$

If $f(\alpha) = 0$, then:

$y_p = x \frac{1}{f'(D)} e^{\alpha x} = x \frac{1}{f'(\alpha)} e^{\alpha x}$, provided $f'(\alpha) \neq 0$

If $f'(\alpha) = 0$, then:

$y_p = x^2 \frac{1}{f''(D)} e^{\alpha x} = x^2 \frac{1}{f''(\alpha)} e^{\alpha x}$, provided $f''(\alpha) \neq 0$

and so on…

Problem 1. Find the general solution of: $y'' + 5y' + 4y = 18e^{2x}$

Solution: The given equation is:

$$y'' + 5y' + 4y = 18e^{2x} \qquad (1)$$

S.F. : $(D^2 + 5D + 4)y = 18e^{2x}$ where $D \equiv \frac{d}{dx}$

$\Rightarrow f(D)y = r(x)$ where $f(D) = (D^2 + 5D + 4)$ and $r(x) = 18e^{2x}$

To find Complimentary Function (C.F.):

A.E. : $f(D) = 0$ $\Rightarrow (D^2 + 5D + 4) = 0$ $\Rightarrow (D + 1)(D + 4) = 0$

$\Rightarrow D = -1, -4$ (real and unequal roots)

Let $m_1 = -1$ and $m_2 = -4$

$\therefore$ Complimentary function is given by:

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

$$\Rightarrow y_c = c_1 e^{-1x} + c_2 e^{-4x}$$

To find Particular Integral (P.I.):

$$y_p = \frac{1}{f(D)} r(x) = \frac{1}{(D^2+5D+4)} (18e^{2x})$$

$$\Rightarrow y_p = 18\left[\frac{1}{(D^2+5D+4)} e^{2x}\right]$$

$$\Rightarrow y_p = 18\left[\frac{1}{\left((2)^2+5(2)+4\right)} e^{2x}\right] \qquad (Put\ D = 2)$$

$$\Rightarrow y_p = 18\left[\frac{1}{18} e^{2x}\right] \qquad \Rightarrow y_p = e^{2x}$$

$\therefore$ General solution is given by: $y =$ C.F. +P.I.

i.e. $y = y_c + y_p$

$\Rightarrow y = (c_1e^{-1x} + c_2e^{-4x}) + e^{2x}$ **Answer.**